# Reasoning Trace Validation via $\ell^1$ Obstruction Geometry

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**Abstract**

This paper provides a formal finite obstruction protocol for the **auditability and validation of reasoning traces**. We establish that textual reasoning outputs from large language models (LLMs) or automated toolchains cannot be scored or validated directly via language-level heuristics or embedding similarity. Instead, we define a **realization boundary** where raw text is mapped to a finite structural carrier $(P, \omega_1, \mathcal{A}, \mathcal{C}, \tau)$ before validation occurs. By enforcing a **replayable certification** requirement and a **typed refusal taxonomy**, we provide a framework for reasoning-validity claims that is structurally isolated from rhetorical plausibility.

**The Core Doctrine.**

| The verifier does not judge language directly. It judges declared finite structure. |
|---|

## 1 Scope: Interfacing with Modern AI Verification

Earlier papers in this series establish the abstract algebra of obstruction (Paper 009) and the governance of evaluators (Paper 011). This paper operationalizes those theories to solve the concrete problem of **trace validation**. It provides the first formal interface between raw textual reasoning and audit-grade certification.

### Operational Context

Traditional AI evaluation relies on scalar reward models, embedding similarity, or "chain-of-thought" scoring—all of which are vulnerable to **shortcut certification** and **reward-model collapse**. This paper replaces these heuristic judgments with:

- **Declared Carriers:** Explicitly defined claim/citation/tool graphs.
- **Replayable Certification:** Proof-carrying evidence that a reasoning transition is exact within the carrier.
- **Typed Refusals:** Informative failure modes (e.g., `CarrierFailure`, `BudgetUnknown`) that provide authority semantics instead of binary outcomes.

This paper does not define thinking, intelligence, or semantic understanding. It defines the structural requirements for a trace to serve as **admissible evidence** for a reasoning transition.

## 2 Realization Boundary

A submitted output $y \in \mathsf{Text}$ is source material. It becomes auditable only after a deterministic or versioned extractor $\pi$ emits a finite validation object

$$V_\pi(y) = (P, C^0, C^1, C^2, d_0, d_1, \omega_1, \mathcal{A}, \mathcal{C}, \tau_\pi).$$

The components are:

| Component | Finite meaning |
|---|---|
| $P$ | carrier graph of claims, citations, tool states, or proof obligations |
| $C^0$ | admissible repairs, expansions, alignments, or bridge steps |
| $C^1$ | observable distinction coordinates |
| $C^2$ | closure and dependency-obstruction coordinates |
| $d_0$ | repair-to-distinction map |
| $d_1$ | distinction-to-closure map |
| $\omega_1$ | submitted edge defect or support-demand cochain |
| $\mathcal{A}$ | admissible repair/operator set |
| $\mathcal{C}$ | admissible context family |
| $\tau_\pi$ | replayable extraction log |

No extraction means no certificate:

$$\pi(y) = \bot \quad \Rightarrow \quad \mathsf{Status}(y) = \mathsf{NeedsRealizationFunctor}.$$

## 3 Trace Carrier Types

The protocol permits multiple finite carriers, provided the carrier is declared before validation:

| Carrier | Vertices | Edges | Typical obstruction |
|---|---|---|---|
| claim graph | atomic claims | entails, contradicts, supports | unsupported or contradictory claim |
| citation graph | cited propositions | citation-support relation | citation does not support claim |
| tool trace graph | tool states | call, parse, transform, commit | output mismatch or hidden mutation |
| proof-obligation graph | obligations | discharge, depends-on | obligation skipped |
| reasoning trace graph | step states | inference, context carry | non-replayable transition |

The same text may realize into several carriers. A validation result is always carrier-relative:

$$\mathsf{Valid}_{\pi,P,\mathcal{A},\mathcal{C}}(y),$$

not unqualified validity.

## 4 Submitted Defect and Repairable Mass

Given $\omega_1 \in C^1$, the repairable component is the part lying in

$$B^1 = \mathrm{im}(d_0).$$

The irreducible quotient obstruction is

$$[\omega_1] \in Q^1 = C^1/B^1.$$

The submitted $\ell^1$ defect is

$$I(\omega_1) = \|\omega_1\|_1.$$

The exact repair problem is

$$\min_{\alpha \in C^0} \|\omega_1 - d_0\alpha\|_1.$$

Define

$$\Phi(\omega_1) = \inf_{\alpha \in C^0} \|\omega_1 - d_0\alpha\|_1.$$

Then:

| Case | Status |
|---|---|
| $\Phi = 0$ and closure holds | exact or certified bridge |
| $\Phi > 0$ and closure holds | cohomological obstruction |
| $d_1\omega_1 \neq 0$ | closure failure |
| extractor lacks coordinates | carrier failure |
| repair exists only outside $\mathcal{A}$ | admissibility failure |
| search budget exhausted | budget unknown |

Skipped steps are not automatically invalid. They are valid only if a replayable $\alpha$ certifies that the skipped bridge lies in $B^1$.

## 5 Dual Witnesses

A dual witness is a finite cochain $\varphi \in (C^1)^*$ satisfying

$$d_0^*\varphi = 0, \qquad \|\varphi\|_\infty \leq 1.$$

Its value is a certified lower bound for the residual norm:

$$\langle \varphi, \omega_1 \rangle \leq \Phi(\omega_1).$$

For an optimal sign-aligned witness, equality certifies the obstruction value.

In a trace audit, a witness localizes irreducible defect:

1. an unsupported edge,
2. a contradiction cycle,
3. a citation-support mismatch,
4. a tool-output mismatch,
5. a dependency path that cannot be replayed under the declared carrier.

The verifier should emit the witness, not a psychological label.

## 6 Closure

Exactness is insufficient if the trace opens a higher obstruction. Closure is measured by

$$\Gamma(\omega_1) = \|d_1\omega_1\|_1.$$

A local $\Phi$-decrease is not accepted when it worsens closure:

$$\Phi(S') < \Phi(S) \quad \text{but} \quad \Gamma(S') > \Gamma(S) \quad \Rightarrow \quad \mathsf{NoPromotion}.$$

The validator therefore ranks proposed repairs by the evaluator law

$$R = (\Phi, \Gamma, -\Xi, \mathsf{NA}, \mathsf{RA}),$$

not by scalar loss or rhetorical plausibility.

## 7 Observer Passivity: The Anti-Shortcut Principle

A common failure mode in modern AI verification is the use of **compressed evaluators**—models that map a complex trace to a smaller diagnostic view (e.g., a "score" or a "summary") before certification.

**Definition (Observer).** An observer is a mapping $\mathcal{O} : C^1 \to C^1_{\mathcal{O}}$ that restricts the coordinate space of the defect cochain.

**Theorem (Observer Passivity / Anti-Shortcut).** For any observer $\mathcal{O}$ that lacks source-level certification:

$$\Phi_{\mathcal{O}}(\mathcal{O}\omega_1) \leq \Phi(\omega_1).$$

The observer can only **weaken** the obstruction; it cannot create certification by erasing coordinates. A boolean judgment of `valid` from a compressed observer is strictly **not certificate-grade** unless it links directly to a replayable witness in the source carrier.

**Consequence.** Scalar loss, reward models, and embedding agreement guide *proposal search* but are structurally incapable of replacing the **finite verifier**. Any certification that bypasses the realization boundary is rejected as a "hidden observer erasure."

## 8 Typed Statuses

| Status | Condition |
|---|---|
| `Exact` | $\Phi = 0, \Gamma = 0$, certificate replay succeeds |
| `RepairableGap` | missing bridge is supplied by replayable $\alpha$ |
| `OpenObstruction` | $\Gamma > 0$ |
| `CohomologicalObstruction` | $\Gamma = 0$, $\Phi > 0$ under the declared repair complex |
| `MissingData` | required evidence edge absent |
| `EvidenceBoundaryFailed` | evidence edge exists but fails support map |
| `ToolTraceMismatch` | declared tool output does not replay |
| `AdmissibilityFailure` | repair requires operator outside $\mathcal{A}$ |
| `CarrierFailure` | current carrier cannot represent required distinction |
| `BudgetUnknown` | search ended without proof of exactness or obstruction |
| `NeedsRealizationFunctor` | no valid extractor supplied |

These statuses are refusal-compatible. The system must not silently promote an untyped residual.

## 9 Validation Algorithm

Input:

$$(y, \pi, \mathcal{A}, \mathcal{C}, B)$$

where $B$ is a finite validation budget.

Procedure:

1. Extract $V_\pi(y)$. If extraction fails, emit `NeedsRealizationFunctor`.
2. Check carrier finiteness and operator declarations. If absent, emit `CarrierFailure`.
3. Compute $\omega_1$, $d_0$, $d_1$, and extraction log $\tau_\pi$.
4. Solve or bounded-search the exact repair problem.
5. Compute $\Phi$, $\Gamma$, and available witnesses.
6. Replay any proposed repair $\alpha$.
7. Emit typed status and certificate hash.

Output:

$$(\mathsf{Status}, \Phi, \Gamma, \varphi, \alpha, \tau_\pi, \mathsf{CertHash}).$$

## 10 Certificate Format

A trace certificate contains:

```
ExtractorId
ExtractorVersion
CarrierHash
OperatorSetHash
ContextFamilyHash
SubmittedDefectHash
RepairWitnessHash | DualWitnessHash | ClosureWitnessHash
ReplayLogHash
Budget
Status
```

The certificate must be replayable. If the extractor, operator set, or context family changes, the certificate is invalid unless a licensed lift preserves the old quotient structure.

## 11 Explicit Rejection of Cognition Claims (Non-Claims)

The structural integrity of this framework relies on the rejection of anthropomorphic equivalences. We explicitly state:

- **Trace Text $\not\Rightarrow$ Truth:** Textual output is source material, not a direct witness of truth.
- **Reasoning Trace $\not\Rightarrow$ Reasoning:** Admissibility of a trace implies structural replayability, not the presence of "thought."
- **Stream Projection $\not\Rightarrow$ Inner Voice:** The "chain-of-thought" is a serial emission, not a window into internal cognition or consciousness.
- **Low Scalar Loss $\not\Rightarrow$ Valid Trace:** Numerical optimization is not a proxy for structural exactness.
- **Embedding Agreement $\not\Rightarrow$ Exactness:** Semantic similarity does not satisfy the requirement for a replayable $\ell^1$ witness.

All valid claims are **finite, carrier-relative, and replay-relative**. We reject any interpretation of this work that maps cochain obstruction to cognitive or subjective states.

## 12 Theorem Targets

**Theorem 14.1, Certificate Soundness.** If the verifier emits `Exact`, then there exists a replayable repair witness $\alpha \in C^0$ such that $d_0\alpha = \omega_1$ and $d_1\omega_1 = 0$ under the declared carrier.

**Theorem 14.2, Witness Soundness.** If the verifier emits `CohomologicalObstruction` with a certified optimal dual witness $\varphi$, then no admissible same-carrier repair in $\mathcal{A}$ eliminates the certified residual.

**Theorem 14.3, Observer Non-Certification.** If an observer $\mathcal{O}$ collapses obstruction-distinct traces, then a downstream scalar or boolean judgment on $\mathcal{O}\omega_1$ cannot certify source-level exactness without an additional replayable lift.

**Theorem 14.4, No Silent Promotion.** A proposed trace repair cannot be promoted if it decreases $\Phi$ while increasing $\Gamma$, violates rollback compatibility, or changes the carrier without licensed lift.

## 13 Final Law

| |
|---|
| A submitted trace is valid evidence only after finite extraction, exact repair or obstruction witnessing, closure check, and replayable certification. |

The verifier does not judge language directly. It judges declared finite structure.

## A Implementation Registry

The following modules in the `omega_engine` implement the trace validation protocol:

- **Trace Extraction:** `trace_realization.rs`
- **Validation Algorithms:** `trace_validator.rs`
- **Certification Format:** `certificate.rs`

## References


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2. Carroll, J. H. *General Theory of Finite Descent Authority*. 2026. Zenodo. https://zenodo.org/records/20195157